

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Third Semester of B. Tech. Examination (EE)

April/May 2012

EE 201 Control Engineering

Date: 03.05.2012, Thursday

Time: 01:30 p.m. To 04:30 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) What do you mean by closed loop control system? Give examples of closed loop control system. What are the advantages and disadvantages of closed loop systems? [05]
- (b) Give the comparison of Block diagram and Signal Flow graph Method. [02]
- Q - 2 (a) Using Block diagram reduction technique for the system shown in fig.I, find the transfer function of the system. [05]

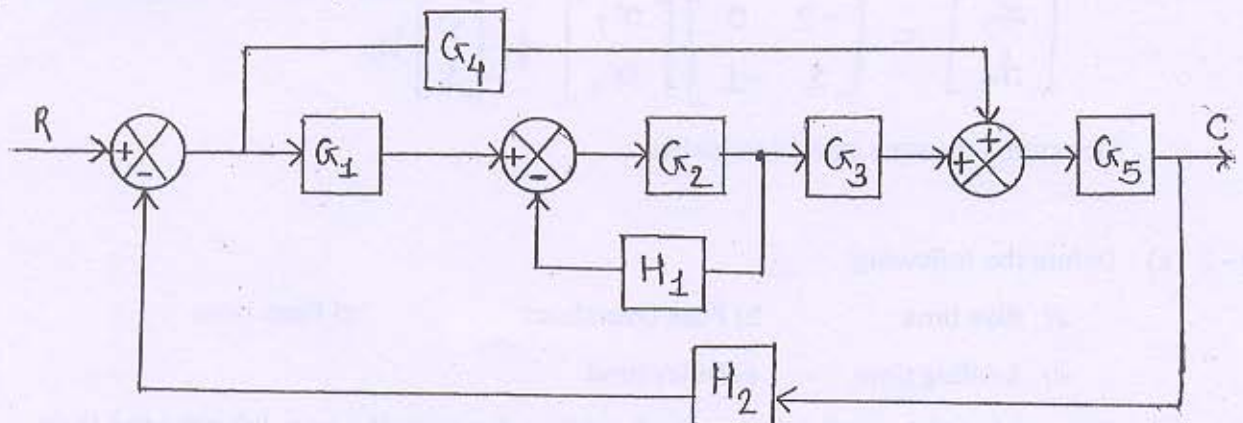


Fig:1

- (b) Find transfer function of the system shown in fig.2 using Mason's gain formula. [05]

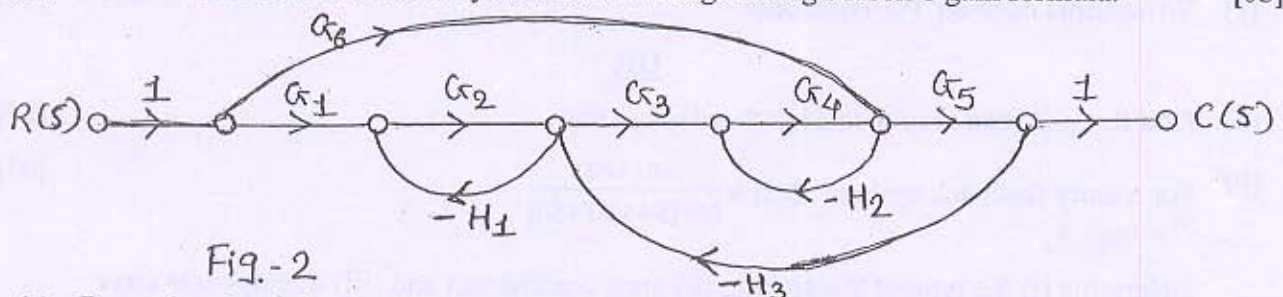


Fig.- 2

- (c) Draw the signal flow graph for the following equations : [04]

(i) $y_2 = a_1 \frac{dy_1}{dt}$ (ii) $y_3 = \frac{d^2 y_2}{dt^2} + \frac{dy_1}{dt} - y_1$

OR

Q - 2 (a) Find the controllability of the system described by the state equation.

[05]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ -2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 3 \end{bmatrix} u$$

(b) The dynamic equation of a system is represented by

[05]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -3 & 0 \\ 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 4 \\ 1 \end{bmatrix} u$$

$$y = \begin{bmatrix} 2 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

Is the system observable?

(c) The state equation of a linear time-invariant system is given below :

[04]

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = \begin{bmatrix} -2 & 0 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u$$

Determine the state transition matrix.

Q - 3 (a) Define the following:

[05]

- a) Rise time b) Peak Overshoot c) Peak time
d) Settling time e) Delay time

(b) State and explain routh's criterion of stability also state the two difficulty and their solution.

[05]

(c) Write short note on PD controller

[04]

OR

Q - 3 (a) Find the unit step response of first order system.

[05]

(b) For a unity feedback system $G(S) = \frac{20(S+2)}{[S^2(S+1)(S+5)]}$

[05]

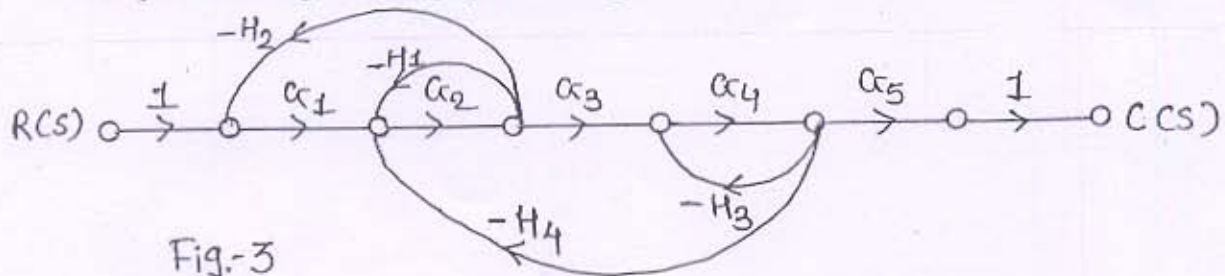
Determine (i) the type of the system, (ii) error coefficients and (iii) Steady-state error for input $1+3t+\frac{t^2}{2}$.

(c) Write the expressions for T_p , T_r , $\%Mp$, T_s .

[04]

SECTION - II

- Q - 4 (a) Determine the number of forward paths and the total number of two non-touching loops for the signal flow graph in the fig. 3. [04]



- (b) Write the expression transfer function, resolvent matrix and state transition matrix. [03]
- Q - 5 (a) The open loop transfer function of a unity feedback control system is [07]

$$G(S) = \frac{K}{S(S+4)(S+6)}$$

Draw the root locus of the system.

- (b) Find the range of K to make the system stable for the unity feedback system using R-H criterion : $G(S) = \frac{K(S+20)}{S(S+2)(S+3)}$ [07]

OR

- Q - 5 (a) Find out the stability of the system by Routh criterion if the characteristic equation is given by $S^4 + S^3 + 2S^2 + 2S + 3 = 0$ [07]
- (b) Sketch the root locus of a unity feedback control system with $G(S) = \frac{K}{S(S+1)(S+3)}$ [07]

And determine the value of k for marginal stability.

- Q - 6 (a) For the system having the open loop transfer function $G(S)H(S) = \frac{10}{S(S+1)(S+10)}$ [07]

Determine the stability of the system by plotting the bode plot of the system.

- (b) Draw bode plot for following and determine gain and phase cross-over frequency. [07]
- Also determine gain and phase margin. $G(S)H(S) = \frac{10}{S(0.1S+1)}$

OR

- Q - 6 (a) Using Nyquist plot, comment on the stability of the following system [07]

$$G(S)H(S) = \frac{60}{(S+4)(S^2+2S+5)}$$

- (b) Decide the stability using Nyquist Plot. $G(S)H(S) = \frac{1}{(S+1)}$ [07]
